

Tag & Track



Conner Museum of Vertebrates

For: Jessica Tir and Katherine Corn

Developers: Martin Hundrup, Shawn Will, and Sarah Carley



Introduction to Tag & Track

- **Objective:** Our objective is to make an iPad app that allows the employees at the Conner Museum to more easily perform their loan tracking.
- **Purpose:** The current way the Museum does things is all on paper, which can get hard to keep track of as well as tedious to work with. Our app aims to make this part of their job less tedious.
- **Scope:** The scope for this project includes loading specimen from arctos, allowing the creating and saving of loans, a way to track what shelf specimen are on, a way to export loan data, as well as a centralized QR code system to make everything quickly scannable.



Team Introduction

- Introductions - Tag and Track Team
 - Martin Hundrup - Team Lead and CS Major, Back End
 - Sarah Carley - CS Major, Front End
 - Shawn Will - CS Major, Database
- Objectives
 - Improve front end loading times and view.
 - Improve database workflow.
 - Implement user login.



Sprint Objectives

Defined Objectives

- Improve User Interface.
- Improve emailing to display more information.
- Improve backend processes.

Rationale

- These components help make the application more efficient for the user and sets up what we will need for future Arctos integration.

Connection to Overall Vision

- These improvements will directly assist the user in the loan creation and viewing process.



Feature Implementation and Demo (Martin)

Feature Overview

- Containers
- Login

Testing and Validation

- Functional



Feature Implementation and Demo (Martin)

TagAndTrack

Employee Login

Select your name or enter a new one:

-- Select Employee --
hundmar

— OR —

hundmar


Login



Feature Implementation and Demo (Martin)

All Containers



Add Container

Cabinet A-1 Main storage cabinet Contains 5 specimen(s)	View
Cabinet A-2 Secondary storage Contains 5 specimen(s)	View
Cabinet B-1 Overflow storage Contains 3 specimen(s)	View



Feature Implementation and Demo (Martin)

Cabinet A-1

Main storage cabinet




Add Specimen to Container

Specimens in this container:

Salmon Specimen

Remove

Lion Claw

Remove

Bear Femur

Remove



Feature Implementation (Sarah)

Feature Overview

- Improved login page after it was implemented.
- Improved data table significantly.
 - Data loads at least 40% faster.
 - Data table more visually appealing while preventing memory leaks.
 - Data table now searchable.

Testing and Validation

- Visible validation was used as well as validation on the iPad (as some UI elements can look different on iPad than when tested on Windows).



Feature Demonstration (Sarah)

Data Table Before:

				Search
Description	Borrower	Email	Date Checked Out	
Bird wing analysis loan	Dr. Lydia Finch	lydia.finch@wsu.edu	11/3/2025 12:00:00 AM	
Fish skeleton research loan	Marcus Lee	marcus.lee@wsu.edu	11/2/2025 12:00:00 AM	
Mammal bone comparison	Claire O'Hara	claire.ohara@wsu.edu	11/1/2025 12:00:00 AM	
Avian feather pigment study	Dr. Tom Rivers	tom.rivers@wsu.edu	11/3/2025 12:00:00 AM	

The old data table had long loading speeds and was not very visually appealing. The search bar was also nonfunctional.



Feature Demonstration (Sarah)

Data Table After:

Search...						
ID	Name	Borrower	Checked Out	Due Date	Status	View Loan
1	Loan #1	University of Alaska	1/1/2026	4/1/2026	On Loan	
2	Loan #2	Smithsonian Museum	12/22/2025	4/11/2026	On Loan	

The new data table implementation was able to improve on efficiency, visual appeal, and functionality. The search bar is now functional and will filter the table accordingly.



Feature Implementation and Demo (Shawn)

Feature Overview

- Emailer
- Internal Database

Testing and Validation

- Functional

T

tagandtrackwsu@gmail.com

To You

6:13 PM

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[EXTERNAL EMAIL]

Loan Name: Final Test

Loan ID: 7980491941351194748

Loan Description: Testing Finally

Client Name: Shawn Final

Date Checked Out: 2/16/2026 6:13:50 PM

Date Due: 12/31/9999 11:59:59 PM

Items included in loan:

ID	ArctosID	Name	Description
1	ARC-000000	Salmon Specimen	Cleaned bone specimen
2	ARC-000001	Lion Claw	Preserved specimen stored in fluid jar
3	ARC-000002	Bear Femur	Egg sample in container



Kanban Overview and Contributions

Kanban Board



MoM Evidence

MoM



Documentation of Report

Report



Responsible Use of AI

Usage

- AI was used at times during development of this project.

Applications and Improvements

- Referenced at times to assist in coming up with approaches to code problems, improving efficiency of certain code by at least 20%.
- Assisted in understanding UI elements and creation in .NET MAUI, shortening the research and creation process for UI elements by 50%.

Limitations

- We would verify any code made by the AI and do thorough testing to ensure it was sound.

We understand that final decisions and deliverables are our responsibility.



Sprint 4 Achievements and Challenges

What went well

- Data flow improvements.
- Data Table improvements.
- Emailer improvements.

These improvements built off of Sprint 3 and previous sprints and refactored the code to improve the elements.

Challenges Faced

- Obtaining an API key.
 - We are in communication with Arctos and are close to getting the API keys.
- Testing (some changes need to be tested on the iPad)
 - Only one team member has an iPad, when testing is needed, they load it up and verify the changes.
- Personal Schedules.
 - With busy schedules, we tried to have more meetings to dedicate time to the project.



Next Step and Retrospective

Next steps:

- Further improve exporting capabilities
- Add ability to add specimens manually (to loans and potentially to the database)
- Improve the User Interface further with icons and formatting refactors.

Client Feedback:

The client provided no feedback during this sprint. We will continue consulting them when it comes to new features and changes.

Retrospective:

What went well

- We were able to greatly improve much that was built last semester.
- We have a better idea of what we need going forward.

What can be Improved

- Further improvements can be made to our checkout workflow.
- Storing history can be greatly improved.

Sprint 5 Changes

- We would like to change our data approach if we can obtain an API key.



Conclusion

Progress

- Backend and frontend general improvements.
- Emailer progress.
- Arctos API keys are closer to being obtained.

Key Implementations

- Emailer now gives more information when a loan is checked out.
- Data tables are now more dynamic and easy to load.

Future Direction

- Real data integration (with Arctos if possible)
- Exporting loans, specimen information, etc.
- User Interface improvements.

Thank you!